🖥️ How a Motherboard Starts When You Press the Power Button

(From SMPS to CPU – chip-level explanation)

1️⃣ SMPS is NOT fully OFF (Standby Power)

Even when PC is **shut down**, SMPS still gives **one voltage**:

🔌 SMPS Standby Output

Voltage	Name	Purpose
+5VSB	5 Volt Standby	Keeps motherboard alive

➡️ This **5VSB** goes directly to:

- **Super I/O chip**
- **Power button circuit**
- **RTC / CMOS**
- **Embedded Controller (EC)**

📌 **Without 5VSB, power button won't work**

2️⃣ Power Button Pressed → Signal Generated

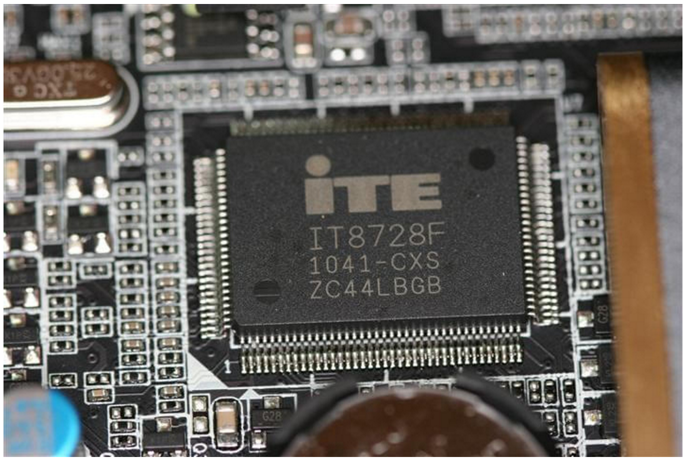
When you press the **Power Button**:

🟢 Signal Path

Power Button

↓

Super I/O / EC Chip



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PS_ON# Signal

- Power button **does NOT power CPU directly**
- It only sends a **logic signal** to Super I/O

🔧 **Important Signal**

Signal	Type	Meaning
PS_ON#	Active LOW	Tells SMPS to start

➡️ # means **active LOW**
➡️ Super I/O pulls **PS_ON# = 0V**

3️⃣ SMPS Turns ON (Main Power Rails)

Once **PS_ON# goes LOW**, SMPS starts fully.

⚡ **SMPS Outputs Now**

Voltage	Used For
+12V	CPU VRM, GPU
+5V	USB, logic
+3.3V	Chipset, BIOS
-12V	Legacy (rare use)

📌 These voltages go to **motherboard power stages**, NOT directly to CPU.

4️⃣ Power Good (PWR_OK) Signal

After SMPS stabilizes voltages:

🟡 **SMPS Sends**

PWR_OK = HIGH (≈5V)

Signal	**Meaning**
PWR_OK	Voltages are stable

➡️ Motherboard now knows:
"Power is safe to start CPU"

5️⃣ Motherboard VRM Starts (Most Important Part)

CPU **cannot run on 12V**.
It needs **~1.0V to 1.4V**, very high current.

⚙️ **VRM (Voltage Regulator Module)**

VRM Converts:

+12V → +1.0V (Vcore)

🔩 **VRM Components**

- PWM Controller IC
- MOSFETs (High & Low side)
- Chokes (Inductors)
- Capacitors

📌 VRM is located **near CPU socket**

6️⃣ CPU Power Rails (Chip-Level)

CPU needs **multiple voltages**, not only one.

🧠 **CPU Power Lines**

Voltage Rail	Purpose
Vcore	CPU cores
VCCSA	System Agent
VCCIO	I/O
VCCGT	Integrated GPU
VCCPLL	Clock

➡️ All generated by **VRM phases**

7️⃣ CPU RESET & CLOCK Signals

Before CPU runs, two critical signals:

⏱️ Clock Signal

- Generated by **Clock Generator / PCH**
- CPU cannot work without clock

🔁 RESET# Signal

Signal	Meaning
RESET# LOW	CPU held
RESET# HIGH	CPU starts

➡️ Reset is released **only when**:

- Power OK
- VRM stable
- Clock present

8️⃣ CPU Starts Execution (First Instruction)

Now CPU is alive 🟢

🧾 What CPU Does First

1. Looks for **BIOS/UEFI**
2. Reads from **SPI Flash chip**
3. Executes **firmware code**
4. Initializes:

    ○ RAM
    ○ Chipset (PCH)
    ○ GPU
    ○ Storage

📌 CPU starts at **fixed address** (Reset Vector)

9️⃣ RAM Initialization

CPU cannot work without RAM.

🔄 **Steps**

- BIOS trains RAM
- Sets voltage & timings
- Enables memory controller

If RAM fails:

- No display
- Beep codes / LED error

🔟 Display Output & POST

Now system reaches **POST**:

- CPU ✔️
- RAM ✔️
- GPU ✔️
- Keyboard ✔️

➡️ Logo appears on screen

➡️ System ready to boot OS

🧩 COMPLETE SIGNAL FLOW (Text Image Diagram)

[Power Button]
    ↓
[Super I/O / EC]

↓ (PS_ON# LOW)
[SMPS]
↓
+12V +5V +3.3V
↓
[VRM Section]
↓ (Vcore ~1V)
[CPU]
↓
[BIOS SPI Chip]
↓
[RAM Init]
↓
[Display / POST]

🧪 Common Board-Level Fault Examples (Real Repair Insight)

Symptom	Likely Issue
Dead board	No 5VSB
Button not working	Super I/O issue
Powers ON then OFF	VRM fault
No display	CPU/RAM/BIOS
Fans spin, no POST	RESET or CLOCK missing